

Multisensory Teleportation in Virtual Reality Applications

Francesco Soave*

Queen Mary University of London

Ildar Farkhatdinov†

Queen Mary University of London

Nick Bryan-Kinns‡

Queen Mary University of London

ABSTRACT

This position paper aims to briefly summarise existing research in vection, teleportation and multisensory stimuli, present our cross-disciplinary research setup and argue towards the importance of discussing haptic feedback design in Virtual Reality (VR) locomotion techniques. In particular, haptic feedback stimulation has been shown to enhance the perception of self-motion when applied to various parts of the body. The recent developments of haptic devices opens the possibilities to explore "whole-body" haptics in virtual environments for locomotion techniques. We argue that crossmodal stimulation frameworks that have been already applied to study self-motion in VR could potentially provide benefits to locomotion studies.

1 BACKGROUND AND EXISTING RESEARCH

In recent years, also thanks to the deployment of commercially available headsets for the general public, research in VR has increasingly oriented towards multisensory interaction and interdisciplinary implementations such as spatialized audio and tactile feedback. Locomotion studies have been conducted to discuss taxonomies, issues and approaches which often include studies on user experience and presence. However there still seems to be a gap in the literature concerning locomotion techniques and multisensory environments. In this section we summarise existing research in locomotion, some multisensory studies on the perception of motion in VR (which looked at sound as well as haptic feedback in VR).

1.1 Vection in Virtual Reality applications

Vection is a condition for which a stimuli (e.g. visual) can induce a strong and embodied sensation of locomotion relative to the fixed surrounding environment, even when the body is not physically moving [28]. The traditional real life example, is the sensation of motion that arises when observing a departing train on a neighboring railway track. It was demonstrated [11] that the environment configuration that provokes vection, also influences visually evoked postural responses (VEPR). Most studies on vection are focused on vision since that's the stimuli that was found to provoke the strongest sensation. The parameters which mostly affects vision-related vection are the speed of the animation, the vision field (i.e. central and/or peripheral vision) and the depth of the scene [20].

The literature on auditorily and haptically induced vection is rather scarce and highly heterogeneous in terms of settings and setups and it is one of the main objects of our research. Generally speaking, most studies present substantial differences in their setups, aims, approach and methodologies and that gives us the flexibility to explore completely new scenarios. For instance, most studies on auditory and tactile vection do not have a visual component and participants simply are blindfolded [7, 8, 27]. A set of interesting studies that explored crossmodal perception with auditory and visually induced vection comes from Seno et al [23]. In their experiments, they found

that perceptually plausible sounds actually increased self-motion illusion. Their visual stimuli was either a radial particle field moving vertically or a horizontal set of lines. The auditory stimulus was a signal of increasing/decreasing intensity or pitch. Their results showed that plausible variations in intensity (accompanied by particle flow) facilitated forward vection while frequency variations (with moving lines) increased reported upward/downward motion illusion. They gave each user a physical button to record the onset and offset time of the self-motion sensation and used a modified version of magnitude estimation (scale 0 - 100) to collect data about the strength of the illusion. In regards to Virtual Reality applications, the importance of vection is well considered by Riecke [21] where they highlights how movements in the real world are normally combined with the sense of self-motion. He suggests that if we were to not consider this aspect, our virtual applications would be severely limited in terms of immersion and presence. In the same paper researchers provides one of the first evidences that haptic (vibrotactile) stimulation can increase visually induced vection. In the setup, they synchronized the vibration with the sound stimulus. In a more recent paper [19], the same author begins by asking if vection can actually be useful in VR. This study suggests that physical motion cues are critical in facilitating the so called "automatic spatial updating" of the self-to-surround relationships during perspective changes and that self-motion illusion could be exploited to reduce the likelihood of "getting lost" in a virtual environments.

1.2 Multisensory stimuli for locomotion

The literature on locomotion and multisensory stimuli in VR seems to be fragmented and it's difficult to find works that comprehensively discuss design frameworks for multisensory locomotion. Locomotion taxonomies and user studies generally compare techniques that are only implemented in the visual domain. They generally also include some correlated sound effects but there doesn't seem to be a strong body of work in that aspect. Even less when discussing haptic feedback. Haptic feedback is often explored through hand/body devices to provide either additional information of the virtual scene or to observe variations in presence and user experience. However to the author's knowledge there is a lack of frameworks to design haptic feedback to provide an enriching locomotion experience. Some studies were carried out on multisensory stimuli for vection, although often not linked to specific types of locomotion. Keshavarz [6] found that auditory stimuli increase compellingness of visual vection and confirmed that self-reported auditory vection happens in about 25-60% of participants. Some studies on auditory-haptic vection do not have a visual component and participants are simply blindfolded [8, 18]. Notably Valjamae [27] did not find haptic feedback to facilitate or increase the intensity of self-reported vection, while Riecke did [18]. Furthermore, the role of vibrotactile stimuli often seems to be just a marginal addition to auditory and/or visual cues. Some researchers [3, 13] explored the influence of haptic feedback applied to the feet of a person. However these studies are hard to compare. Riecke [18] haptic feedback was generated by adding a small USB fan to a hanging hammock chair to produce "barely noticeable" vibrations at 7Hz. Participants were blindfolded and the study focused on circular vection. On the other hand, Farkhatdinov et al. [3] designed a visual optical flow, while Nilsson [13] had standing participants in a realistic visual context (e.g. a train and a lift).

*e-mail: f.soave@qmul.ac.uk

†i.farkhatdinov@qmul.ac.uk

‡e-mail: n.bryan-kinns@qmul.ac.uk

1.3 Teleportation in VR applications

In physics, quantum teleportation (also known as “entanglement”) has been achieved in laboratory. This type of teleportation, is defined as a protocol in which an unknown quantum state of a physical system is measured and then ‘reassembled’ at a different location (although the physical components of the original don’t move from the sending location) [15]. In multimedia, teleportation can be defined as a locomotion technique, a method to move from one place to another. Its main characteristic is that the travel time is very short compared to other methods, and often assumed to be instantaneous (time $t = 0$). Differently from quantum teleportation, this is not possible to achieve in real world. However in Mixed Reality Environments such as VR games and applications, teleportation is one of the most commonly used locomotion techniques.

Recent developments of XR applications along with multisensory input and stimulation (i.e. visual + sound + haptics) led to investigate further locomotion techniques and possibilities that open up. Walking is the most usual form of locomotion for humans and is correlated with various sensory feedback such as visual, somatosensory, vestibular, efference-copy and auditory. For video games and simulators that provide on-screen visual feedback, we generally do not experience serious problems while using keyboards or joysticks to walk around the VR scene with a character. However, it has long been shown that using such interfaces for walking causes more motion sickness, disorientation [22] and reduced immersion [24] compared to interfaces which utilize body-based input from users. One to one mapping of the movements of the user in the real environment to the virtual environment requires a large work space and costly sensing instruments. Using unidirectional treadmills [16] also require expensive hardware and generally suffer having low mechanical time constants to match human movement intentions. Redirection techniques [17] can be utilized but do not guarantee a non-interrupted VR experience. Walking-in-place systems which use walking-like periodic gestures like in-place marching [4, 29], foot-sliding [26] and arms swinging [14] trade-off between the naturalness and simplicity of the interface.

Point and teleport technique: One way to work around the additional sensory requirements of locomotion in VR is by teleporting [1] rather than walking. This introduces a new modality for locomotion that people are not used to exercising in real life and are only familiar with it through science fiction literature, yet simplifies the locomotion problem to only require a manual controller.

2 EXPERIMENTAL SETUP AND POSSIBLE APPLICATIONS

This paper aims to discuss the opportunity for haptic feedback technologies to address the design of locomotion effects such as teleportation. Research in locomotion in XR applications often compares different methodologies, but to the author’s knowledge, there is no literature on haptic feedback design aiming at establishing a framework for teleportation effects in virtual environments. By observing how teleportation and scene transition have been applied in other media, we aim to open a discussion on the design of crossmodally congruent haptic feedback. We argue that locomotion techniques should also be discussed and implemented considering multisensory experiences. The development in vection research could be linked to research in locomotion and haptic feedback. Perhaps research on locomotion could draw inspiration from collateral research on crossmodal correspondences where stimuli in different domains influence each other. Studies on teleportation in VR [5, 10, 12] often discuss scene transitions analyzing how it has been accomplished in other media such as cinema, so perhaps a similar approach could be undertaken in the design of haptic effects to enhance the locomotion activity (i.e. analyzing technical properties of visual/audio effects, extract the core features and find related representation in tactile domain). The non-realism of teleportation allows researchers and developers to explore new techniques but also investigate the

expectations of users for a multisensory experience. Our experimental setup is concerned with producing a so called “whole-body” vibration in a seated VR application [25]. The system is made of 8 actuators to provide vibrotactile haptic feedback to the body of a person in a VR environment. Our preliminary results showed a positive response to haptic from the users and an increase perception of motion, however questions remains on whether it is possible to identify standard design frameworks for multisensory stimuli applied to locomotion techniques in VR. Research on full body haptics often focuses on audio-haptic cross modal mappings and multisensorial integration, which is regarded as a key point on building consistent VR experiences [2]. For example, Lindeman et al. [9] designed and implemented a wearable suit made of multiple individually addressable vibrotactile actuators placed on the upper body of a user. We are planning to conduct further user studies to analyse the effect of whole-body vibrotactile feedback in a virtual environment when sudden changes in acceleration (velocity spikes) are presented to the user in the different domains (visual, audio, haptic). With the assumption of teleportation as a fast movement between multiple location we expect our findings to be relevant for locomotion techniques such as point to teleport. However, depending on the actual implementation, we believe our research can also bring new insights on other locomotion techniques, in which the motion itself is slower (e.g. flying, floating).

3 PROPOSED DISCUSSION TOPICS

We would like to take the opportunity of the workshop to discuss the following aspects of locomotion:

- Benefits and drawbacks of existing frameworks to design locomotion in VR
- Design frameworks for multisensory teleportation in VR
- Discuss the user expectation of multisensory locomotion

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